IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Digital Object Identifier

Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT After a load is restored or a line is reclosed, residual dynamics can resemble a new intervention. We study sequential attribution from eight PMUs in an IEEE 39-bus model with nine candidate load buses and nine lines. A bank of switched differential-algebraic predictors compares intervention and recovery histories from estimated past states. Conditional Bayesian evidence integrates load severity, while separate active and cause weights retain the explanation of a restored intervention. In an independent simulation evaluation, 108 interventions are replayed under three network conditions and two noise levels. The physical bank produces 644/648 correct family/source decisions before restoration with 72 unmatched alarms, versus 330/648 with 29 for a fixed causal ExtraTrees adapter. The gain concentrates on loads; line coverage matches the short-window learned comparator. A controlled posthoc study identifies a state-carry benefit under both marginal and profiled severity scoring: retaining the state gives 108/108 timely decisions and one extra alarm, compared with 75/108 and 166 after resetting physical deviations. The results establish recovery-consistent prediction as a useful design choice for sparse-PMU attribution and quantify its coverage–alarm trade-off under network uncertainty.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [1]; physical outage detectors compare network hypotheses sequentially [2], [3]; and simulated fault trajectories support diagnosis from sparse PMUs [4]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.
2) A comparison that separates detection, correct source

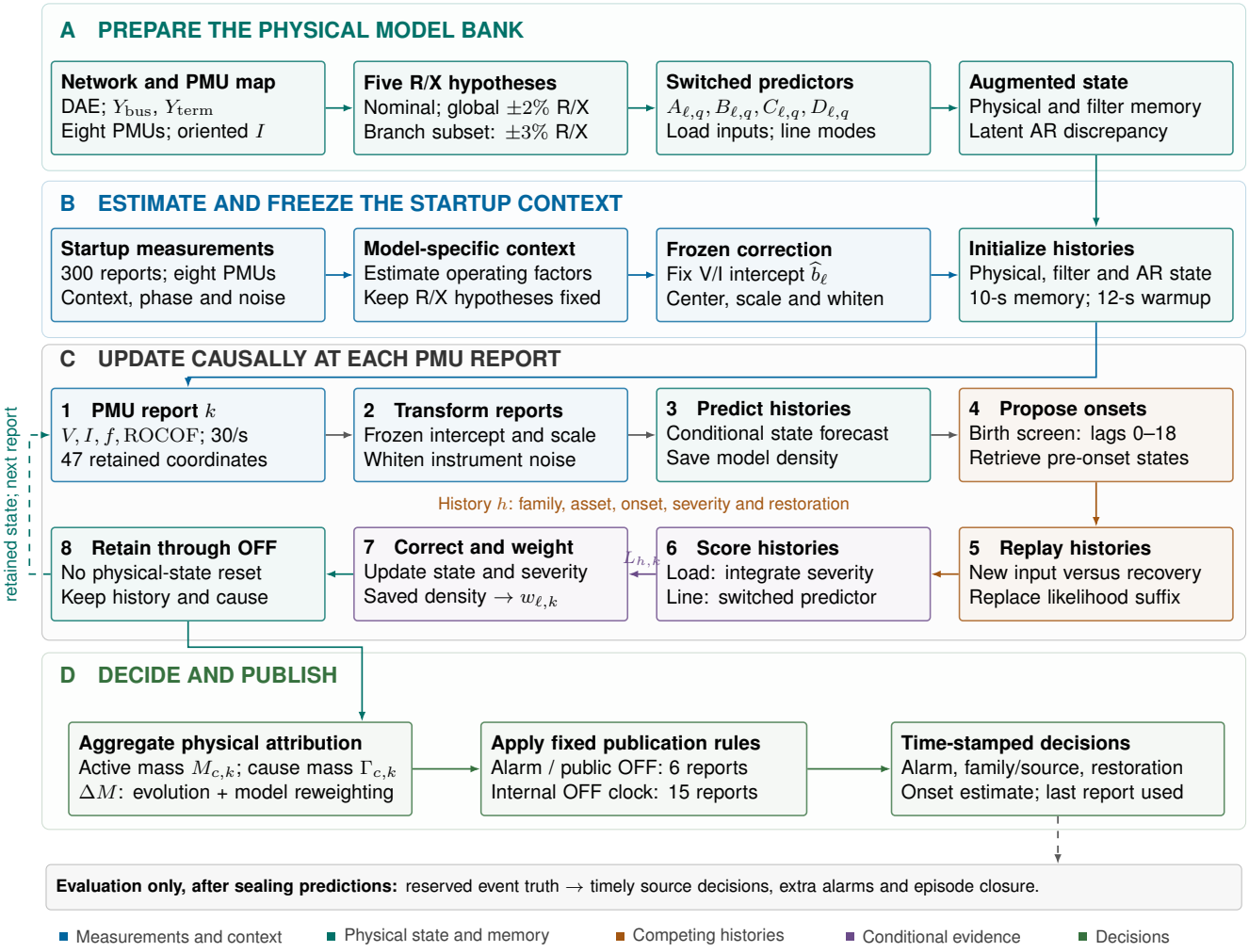


FIGURE 1. Implemented physical-history attribution, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed.

attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.

3) A controlled severity-score/state-carry study on identical records. It quantifies the benefit of preserving physical-state deviations at inferred closure under both marginalization and bounded profiling. The common observations and operators connect the decision improvement to a specific state-retention operation.

The independent comparison gives 644/648 timely attributions with 72 extra alarms. The advantage over the fixed learned adapters concerns load coverage, while network mismatch increases alarm burden. This evaluates configured procedures at their original thresholds. It motivates recovery-consistent inference without establishing superiority at equal false-alarm rates. The supplement documents the implemented filter, complete outcomes and separate context-signature and recovery controls. Earlier voltage, feature and simultaneous-source studies are preserved in the Historical Technical Notes with their own cohorts.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [1], [5], [6]. Functional observability asks whether a requested state quantity can be recovered [7], [8]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [9], [10].

Physical hypotheses support outage inference from phasor angles [11], Bayesian inference with uncertain states [12], and sequential detection under stochastic operation [2], [3].

Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics [13]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [14]–[16].

Recent learned reconstruction also targets sparse sensing. Pomarico *et al.*'s SHRED preprint evaluates dynamic-state reconstruction on IEEE-39 from a limited number of PMUs [17]. Its target is state reconstruction rather than the source/ON/OFF episode contract here. Dey *et al.* detect and localize distribution-network topology changes from phasors using learned statistics, with hardware-in-the-loop validation [18]. Neither sparse sensing nor change localization alone distinguishes the present contribution. These are adjacent tasks, not executed performance baselines in this study.

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [4]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [19]. Multiple-model filtering already propagates conditional state/mode hypotheses [20]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [21]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [22]–[24]. We compare two Extra-Trees models trained for the same 18-source catalog on earlier simulation parents, then frozen with their causal adapters. Their inherited feature families and newly fitted weights are distinguished in the experimental design. Both represented-source learning and a physical dictionary supply source information, through different training and modeling routes. The comparison tests transfer at the original operating points; source-excluded generalization requires a separate experiment.

Physical stochastic models support latent-state estimation [25], [26], while joint DAE parameter inference addresses uncertain generators and networks [27]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [28]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
IMM / M3H	Conditional filters, mode histories and mixture reduction	Hybrid switching-state estimation	Direct history-filter precedent; no common-data baseline here
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

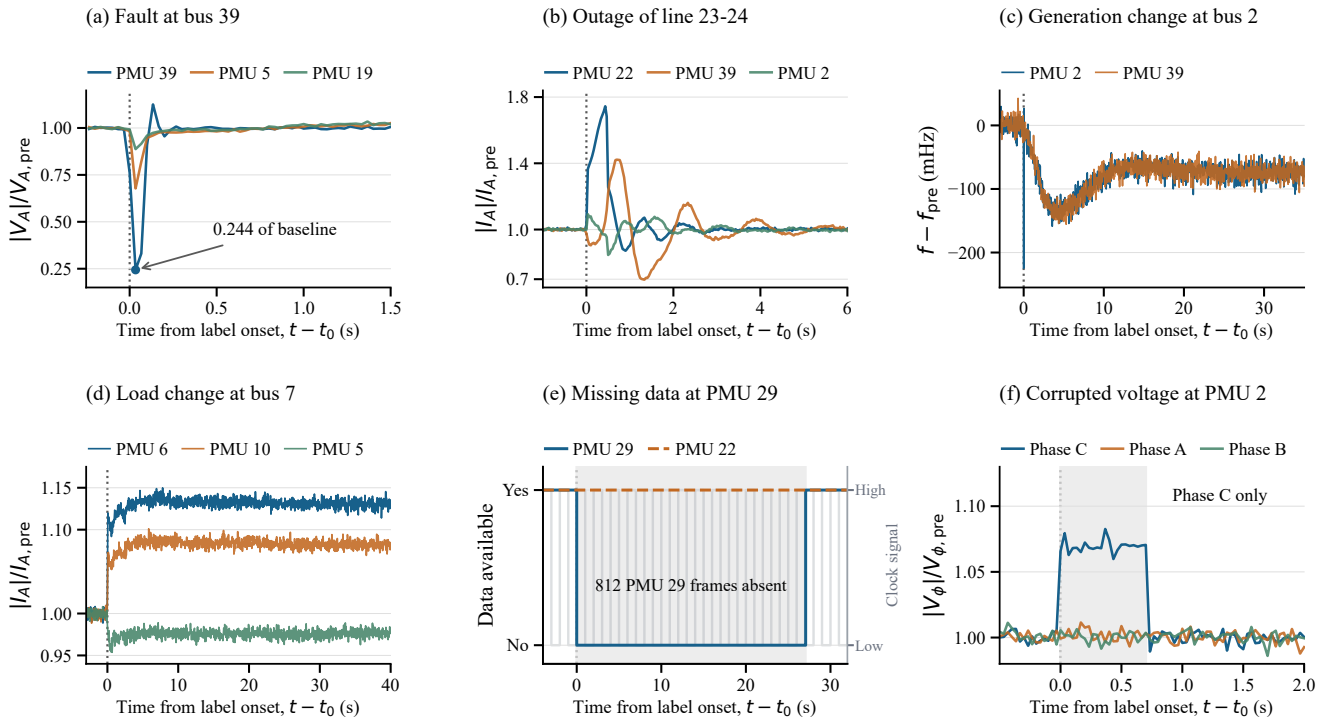


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is applied according to the declared generator contract.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches

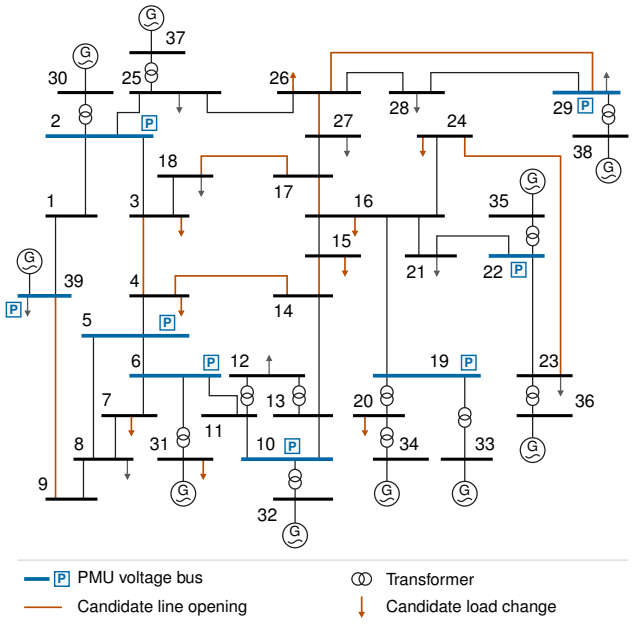


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside the stream test
Unknown	Unresolved diagnosis	Not a new physical family

its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology q, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with a fixed angle reference, the derivatives at that consistent point give, suppressing ℓ,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= c_q + A_q^{red} \delta x + B_q^{red} \delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Network connectivity enters

through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

Exponential discretization integrates c_q over each mode interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section "Implemented Predictor and Conditional Uncertainty" gives the affine construction and its code correspondence.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

The conditional state mean is represented as $m_h + d_h a$. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter v_h. The affine conditional representation retains state–severity dependence:

$$\begin{aligned} \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ \mathrm{Var}(\xi \mid h) &= P_\rho + d_h V_{a,h} d_h^\top. \end{aligned} \tag{14}$$

Conditioning on past reports is implicit. The shared P_ρ comes from nominal-operator Riccati equations, rather than an exact recursion for each switched history. At closure, the filter retains the augmented-state mixture mean and records the amplitude/mean spread that it discards. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

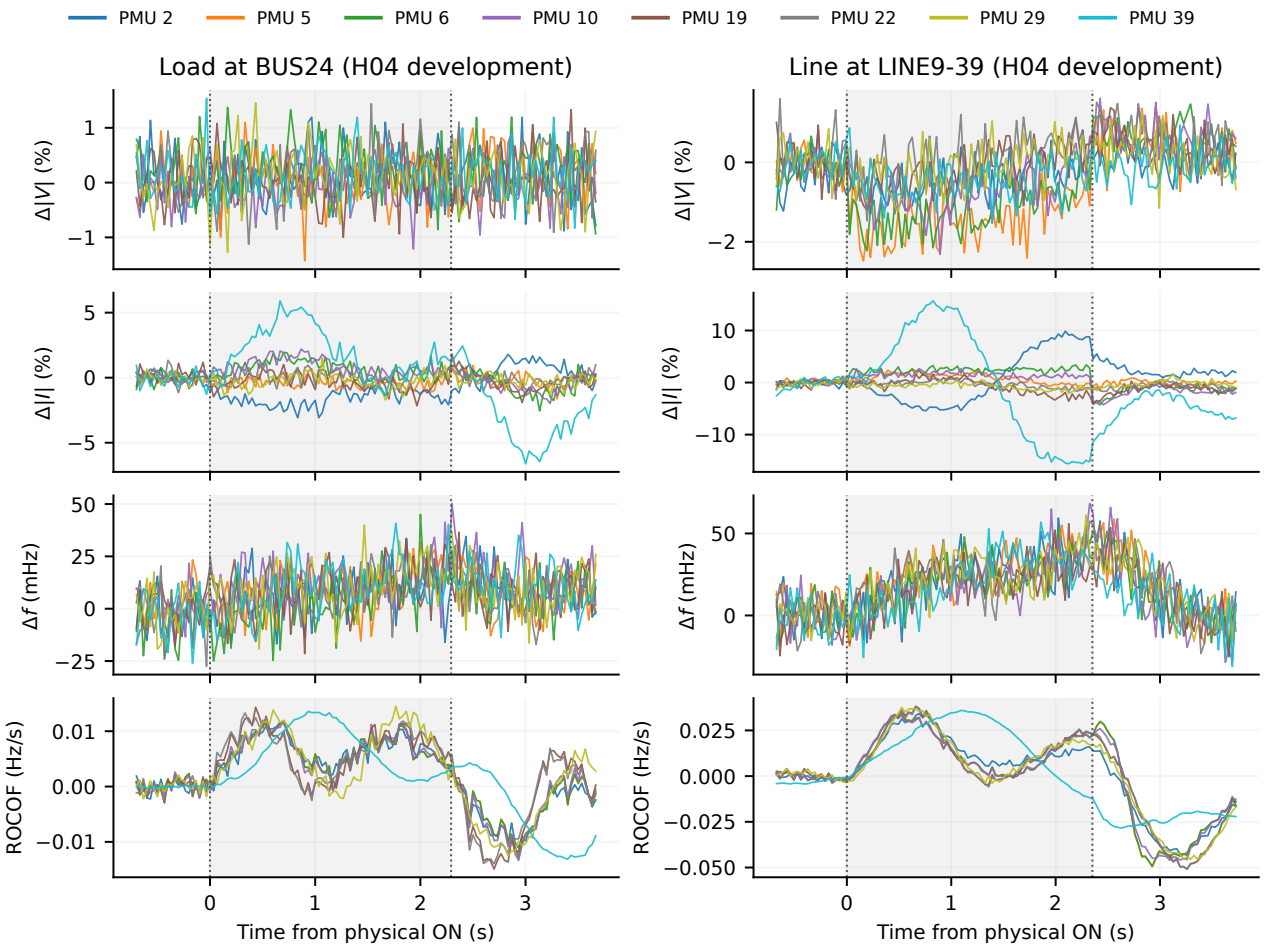


FIGURE 4. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{15}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{16}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_\ell w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{17}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure,

explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \quad (18)$$

The terms distinguish within-model evolution from model reweighting. This identity separates within-model evidence from network-model reweighting. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory ablation.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The reproducibility archive maps these operations to the frozen `transitions`, `process`, and `predictive_log_density` routines and preserves the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h')\pi_{\ell,h',k-1}. \quad (19)$$

IMM filtering approximates mode-conditioned state mixtures; the M3H filter retains mode-sequence hypotheses [20], [21]. Equation (19) is that established mechanism. Here the modes are oriented electrical intervention operators, histories include restoration time, and the output separates active intervention from its originating cause through (17). Retrospective births reuse a stored pre-onset state, while closure compresses histories into a retained augmented-state mean. This specifies an application of history filtering rather than a new general recursion. The state-carry experiment tests one operation in this construction; the preliminary single-state control below requires independent, information-matched validation to identify the incremental benefit of explicit recovery hypotheses.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is

30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The simulator is the IEEE-39 Julia/PowerDynamics research candidate. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

The five-model bank and its nominal version are frozen coreferences. Both ExtraTrees models were trained specifically for the causal load/line catalog, using 32 earlier physical parents for training and eight for development validation. Both training sets include normal operation and all 18 candidate sources. The reference uses 846 causal features from 1-, 5- and 10-s windows; the short-window variant adds 611 features from 3-, 8- and 15-report windows. Each model has 400 trees with the same predeclared hyperparameters. They inherit feature families from the SGSMA project, but their fitted weights are distinct from the original hackathon submission. These models and their adapters were frozen before the present independent schedules; no retraining uses the current evaluation records. All methods receive the same 30-fps observations. The supplementary reproduction table records the training lineage and model hashes.

The learned adapter publishes a change after six consecutive identical argmax labels. Its tree votes are uncalibrated scores. A missing timely diagnosis can therefore reflect the representation, class competition or temporal confirmation; conditional accuracy among emitted diagnoses does not identify which mechanism caused an omission. The fixed-adapter comparison measures transfer of these configured procedures rather than an alarm-budget-matched architecture test.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [2]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

TABLE 3. Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent results are followed by a development example and the controlled component study.

VI. RESULTS AND CONTROLLED COMPARISONS

A. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched event-stream alarms.

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

TABLE 5. Sealed continuous-stream decisions. Each family has 324 replayed instances from 54 distinct interventions (108 across both families); extras are unmatched alarms on event records. The methods retain unequal alarm operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

Its nominal counterpart makes 643/648 with 92 extras. The frozen causal ExtraTrees reference makes 330/648 with 29; its short-window variant makes 352/648 with 16 extras. These are paired replays of 108 interventions, not 648 independent events. Figure 5 shows why the totals differ: the physical banks recover nearly all loads, while both learned adapters already recover most tested lines. The extra alarm burden of the physical banks rises sharply when true R/X changes fall outside their five-model bank.

Figure 6 separates a fast conditional median from coverage: a method can have short delays on its available diagnoses while leaving many load events without a timely source decision.

A posthoc audit of the original episode matches further separates coverage from classification. Every matched diagnosis emitted before OFF has the correct family and source in all four procedures; their errors in the timely endpoint are missing or late diagnoses. The load advantage therefore concerns timely coverage. It does not demonstrate higher conditional classification accuracy among diagnoses already available before OFF. The supplement shows the complete confusion view and counts unmatched alarms separately.

The paired intervals in Fig. 7 confirm a timely-attribution advantage over this ExtraTrees reference in each tested stratum. They also show a higher rate of unmatched alarms under out-of-bank R/X. Those intervals resample physical parents, preserving correlation across noise replays and network conditions. They do not establish superiority at a common false-alarm operating point. The physical bank's stratum median diagnosis delay is about 0.27 s in matched and in-bank conditions; the ExtraTrees medians range from 0.19 to 0.30 s. There is no general latency advantage.

The separate persistent-line test scores the first stop. Both physical banks and both ExtraTrees adapters identify the right line first on 162/162 openings. The declared Chen2016 adapter does so on 74/162; 66 records stop before the opening and 22 stop afterward on a wrong line. A separate descriptive audit of the same persistent records shows 164 and 179 later unmatched alarms for the two ExtraTrees procedures, versus 0 for the physical bank. Correct first identification therefore does not imply stable maintenance of a sustained event. This audit does not change the first-stop comparison or apply a recurrent score to Chen. Chen also stops on all 72 paired normal records, whereas the continuous methods alarm on none. Chen's normal exposure ends at its first stop (2,522 s); the other methods monitor 42,336 s each. These unequal exposures and the adapter's white-injection assumption prevent a general claim against published quickest-change detection. Full stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

The physical dictionary supplies responses for each catalog source, whereas the learned adapters retain their original training and feature maps. The comparison therefore measures transfer of those particular implementations, rather than an information-matched test of physics against supervised learning. It also uses different alarm operating points.

B. COMPUTATIONAL COST

The recorded median history-inference stage exceeds the record duration; its amortized processing rate is below the 30-report/s input cadence. The stage ranges show substantial variation. Per-report latency percentiles, peak memory and a matched-hardware speed comparison were not recorded;

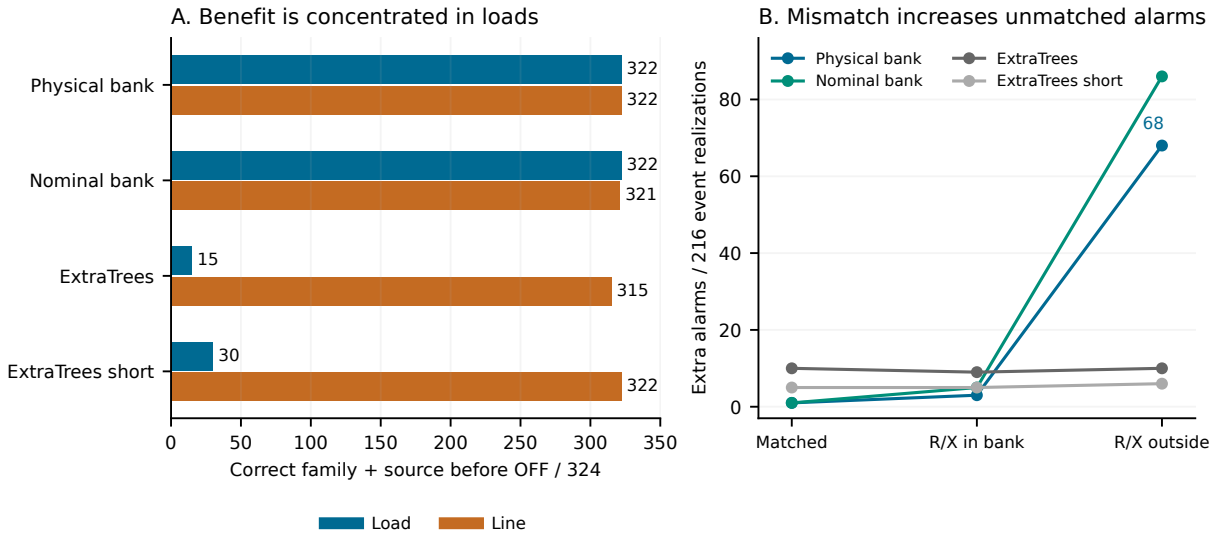


FIGURE 5. **Sealed continuous evaluation. (A) Timely correct family and source by event family, with 324 paired realizations per family. (B) Unmatched event-stream alarms across the two noise levels within each R/X condition, with 216 event realizations per condition. The plots display the gain in load attribution and the alarm cost under out-of-bank network error together. Each method retains its own frozen publication rule and false-alarm budget.**

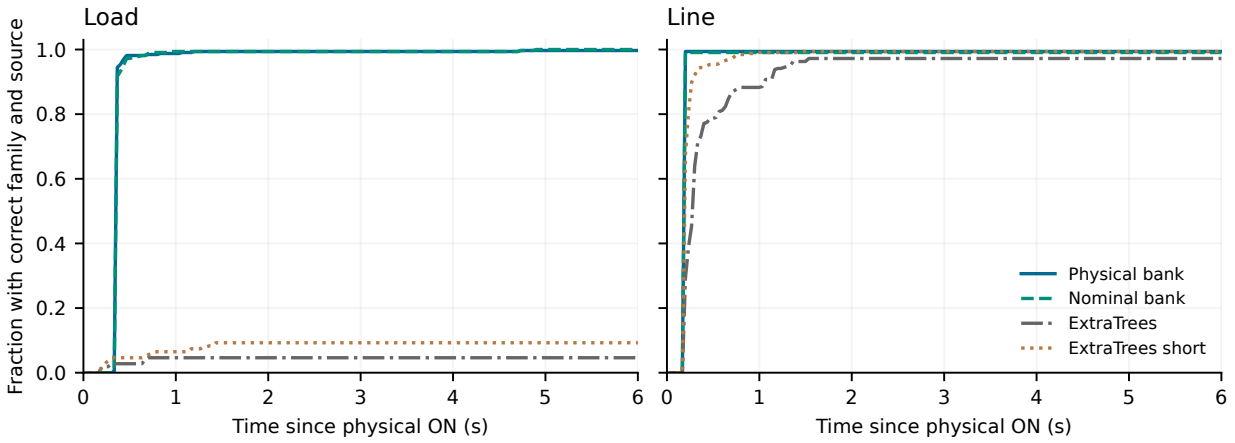


FIGURE 6. **Posthoc descriptive distribution of correct diagnosis time, using all 324 realizations per true family in each denominator. An incorrect or absent diagnosis contributes no cumulative mass. Curves include late correct decisions and are therefore distinct from the before-OFF endpoint; physical OFF varies between events. The correlated replays are not independent observations.**

TABLE 6. **Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.**

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

causal emission timestamps must not be read as wall-clock latency.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A fixed development trajectory illustrates the timing criterion (Fig. 8). The network continues moving after OFF; a later event must be evaluated against that predicted response. Under the tested out-of-bank network error, the bus-31 source is recovered only after restoration. It is correct eventually and fails timely attribution. The full development counts, active-mass trace, ΔM decomposition and failed admission control remain in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

An existing open-development control retains one committed augmented physical state through restoration, rather than resetting to equilibrium. It uses the same nominal conditional plant and measurement operators, profiles severity over short alternatives lasting at most 19 reports, and applies the same

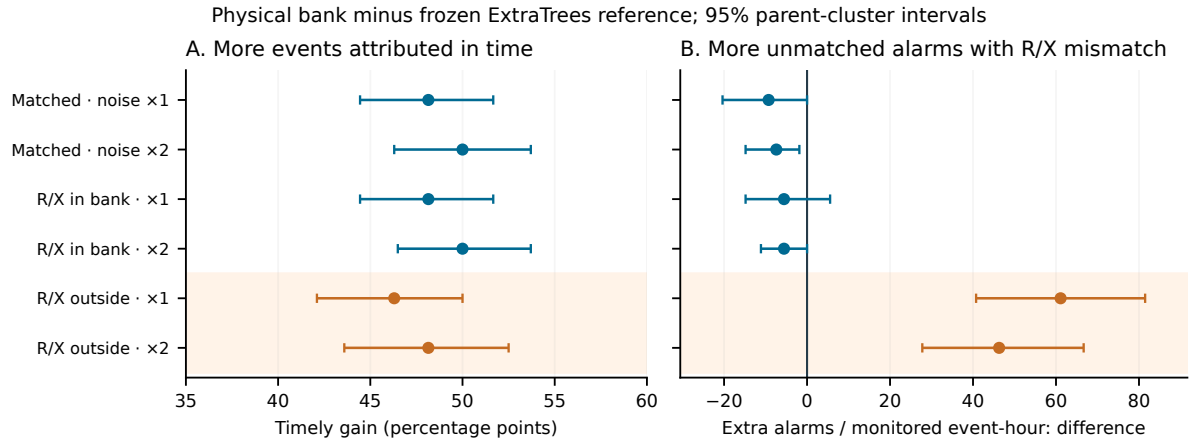


FIGURE 7. **Physical bank minus frozen ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour, excluding the 12-s startup from each 120-s record. The original rate differences and interval bounds are rescaled exactly by 120/108; counts and episode matches are unchanged. The orange rows expose the out-of-bank R/X failure. The intervals describe this finite simulated population.**

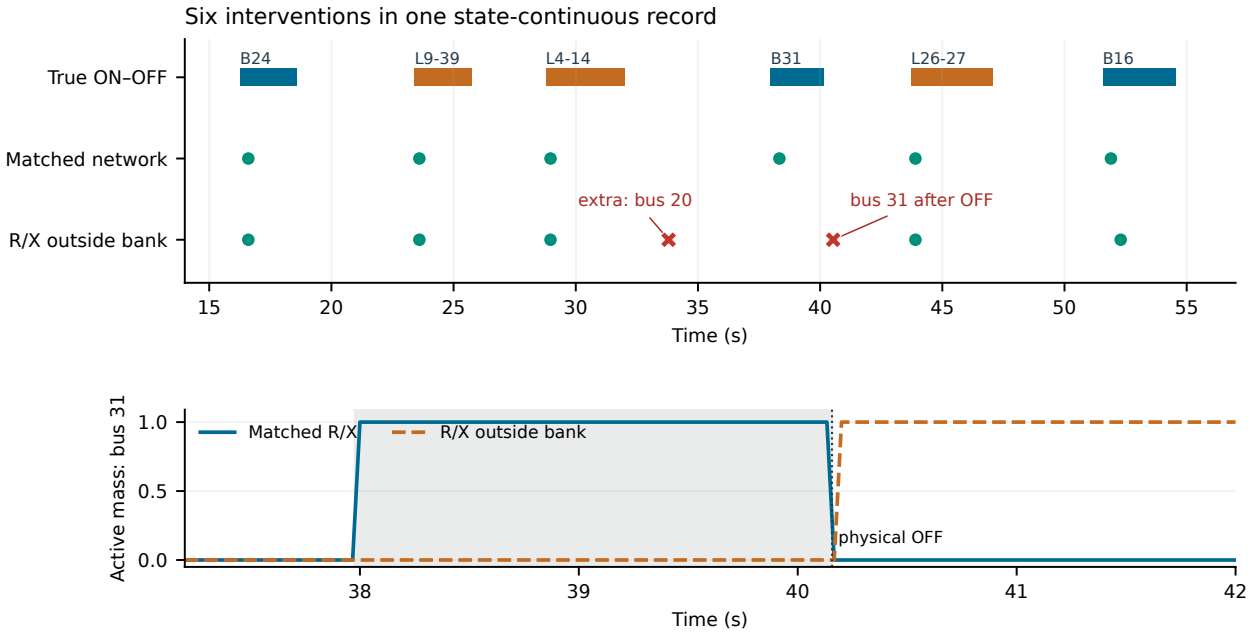


FIGURE 8. **One continuous development trajectory, shown for interpretation rather than selected as an independent test. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass, with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

six-report publication clock to all three procedures. The short GLR achieves 37/54 timely decisions with 37 extras, versus 44/54 and 6 for the nominal history bank, and 51/54 and 3 for the five-model bank. These 54 replays share three development parents; they are separate from the independent test. Different evidence scores, search and severity updates remain, so this is a preliminary continuous-observer comparison, not an isolated causal effect of history retention. Its complete equations and closure outcomes are in the supplement.

E. CONTROLLED SEVERITY-SCORE AND STATE-CARRY STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its

accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{20}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{21}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k/v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 9 report coverage and alarm cost for the four arms. With carry retained, marginal and bounded-profile amplitude scores tie on the reported endpoints. Resetting the first 114 estimated physical-state deviations at inferred closure reduces timely attribution and increases extras under both scores. This reset leaves instrumental state, covariance regime, prior weights, thresholds, clocks and reporting rules intact; simulated states and observations never reset. The profile arm remains a normalized penalized score system and retains conditional Bayesian state estimation. Thus this study supports a physical-state-carry effect at inferred closure in the tested cohort, not the necessity of Bayes or of every form of temporal memory. The supplement reports denominators, normal exposure, persistent episodes, censoring and all paired parent-cluster intervals. These are posthoc within-simulator contrasts at the original thresholds, not a comparison at matched false-alarm rates.

VII. DISCUSSION AND LIMITS

A. PHYSICAL-STATE CONTINUITY AS A DESIGN CHOICE

The strongest controlled result is the state-carry contrast. Physical OFF restores an input or branch, while the machines retain their state. Removing those estimated deviations at inferred closure substantially worsens decisions on the same records. Marginal and bounded-profile severity scores tie on the measured endpoints; both retain conditional state estimation. This supports recovery-consistent prediction, without showing that Bayesian marginalization is necessary for those endpoints.

The history bank gives a concrete way to represent the remaining response and compare it with later interventions. Its complete operational behavior is measured, but the reset control is deliberately severe. A continuous physical observer without explicit recovery-history competition could retain the same consistency. The open-development single-state GLR comparison gives preliminary evidence against one such alternative, with different search and score semantics. Independent validation under matched information and a common calibration contract remains the key unresolved mechanism test.

B. WHAT IMPROVED RELATIVE TO THE LEARNED REFERENCE

The improvement concerns timely coverage of load sources under this known-catalog simulator contract. Line coverage is comparable to the short-window learned adapter. The physical method also issues more extra alarms under network mismatch. The result is therefore a coverage–alarm trade-off at fixed thresholds, rather than dominance at equal alarm rates or a higher official hackathon score. Engineering physical features is useful in the historical learned route; the physical bank additionally computes how each candidate intervention should affect remote PMUs. These routes have different access to source responses and training data.

A matched architecture comparison requires a physical continuous-state control and a history bank with the same operators, together with a learned baseline trained on the permitted catalog. Operating points must be chosen on separate calibration records before a new test. The present comparison preserves the frozen adapters and their failures. The two learned models already saw all candidate classes during their earlier training; their misses cannot be attributed solely to missing source labels or to an unexamined scalar threshold.

C. MODEL CORRESPONDENCE AND MEASUREMENT SCOPE

The nonlinear generator and local predictors share the same research plant family. Switched dynamics, nonlinear responses, noise and out-of-bank R/X challenge the inverse model, but do not remove this shared-model advantage. Independent dynamic-model or cross-simulator transfer remains untested. All nine load and nine line candidates were inherited from method development before the independent draw; they are a declared catalog, not a representative sample of all 39 buses or all event families.

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

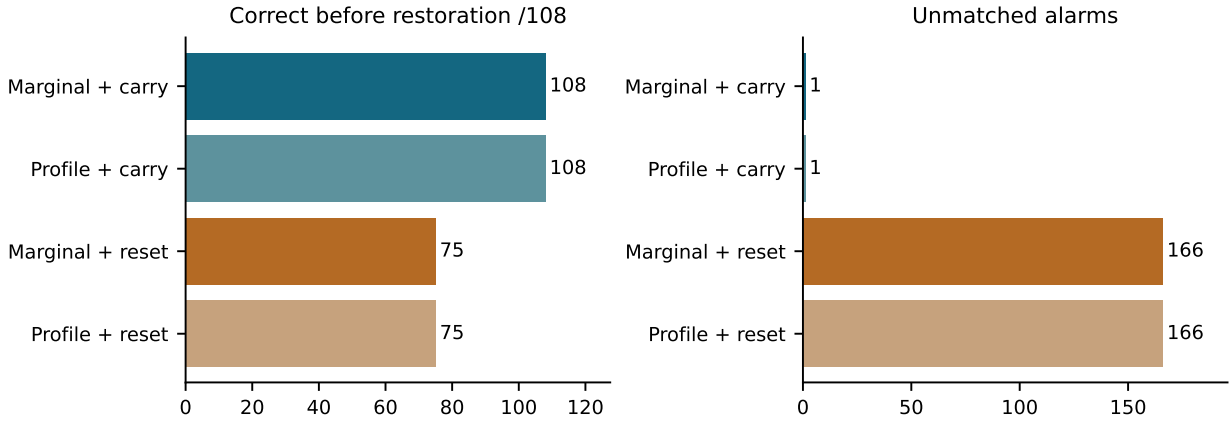


FIGURE 9. Amplitude-scoring and physical-state-carry ablation on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the observed alarm rate may differ between arms.

The eight-PMU terminal map is a research measurement hypothesis. BUS5 retains the provisional 5–6 assignment, and BUS39 current angle is excluded. The synchronized channel grouping includes remote current terminals, so it does not certify an installation of eight colocated voltage/current PMUs. The observation model retains each terminal's actual location; remote voltage–current products are power proxies. The generator applies temporal and cross-PMU noise structure within each signal family, but does not apply its archived cross-family factor covariance. Frequency/ROCOF noise is added after phase estimation. The experiment therefore does not validate a complete physical PMU noise model, a producer's CT placement or exact reproduction of the historical reference record. It also assumes an event-free startup and available PMUs.

D. APPROXIMATION AND DECISION UNCERTAINTY

History pruning, shared regime covariances, plug-in startup estimates and state-mean compression make the bank approximate. Its masses are attribution weights under those approximations, not validated confidence levels. The supplementary numerical checks verify the severity integral, partition identity and conditional-state algebra under their local models. Calibration of the nonlinear stream additionally requires pre-update innovation diagnostics, interval coverage and sensitivity to pruning and state compression. Existing instrumental whitening and latent AR states address distinct components of dependence; their presence alone does not establish predictive adequacy. Discarded mass is recorded by the implementation, while a calibrated reliability and bank-size study remains open. Response fidelity and source discrimination also differ: sources may remain close after fitting severity and context. The supplementary context-signature and recovery-geometry controls preserve their null source-decision results and do not supply an online identifiability certificate. A useful extension would test whether a causal separation margin predicts errors and delays before using it to justify abstention.

The measured research implementation runs below the input cadence. Decision timestamps, computation delay and queueing must remain separate. These results support further diagnostic development, rather than protection or control deployment.

VIII. CONCLUSION

A restored electrical intervention can remain the cause of the observed response. We formulated load/line attribution through physical histories that start from estimated past states and retain that response through restoration. Active and originating-cause weights separate the current intervention from its continuing physical effect.

The independent simulated comparison gives 644/648 timely family/source decisions and 72 extra alarms for the five-model bank, compared with 330/648 and 29 for the fixed causal ExtraTrees reference. The advantage is concentrated in loads; line coverage is similar to the short-window learned comparator. The controlled study identifies a strong state-carry effect, while marginal and profiled severity scores tie. Together, these results support an executable approach that uses electrical dynamics to attribute interventions from remote PMUs, preserves the response across restoration, and

separates timely coverage from alarm burden. Physical-state continuity supplies the consistent reference needed to evaluate subsequent interventions during recovery.

REFERENCES

[1] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[2] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[3] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[4] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023.

[5] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[6] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[7] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.

[8] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.

[9] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[10] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.

[11] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.

[12] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.

[13] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.

[14] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.

[15] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.

[16] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.

[17] A. Pomarico, A. Berizzi, and J. N. Kutz, "A shallow recurrent decoder for dynamic state estimation with a limited number of PMUs in power systems," *arXiv preprint arXiv:2607.00116*, 2026.

[18] A. Dey, S. Chakraborty, and M. V. Salapaka, "Topology change estimation in power distribution networks: A machine learning approach," *IEEE Transactions on Smart Grid*, vol. 17, no. 4, 2026.

[19] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.

[20] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.

[21] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.

[22] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.

[23] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.

[24] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.

[25] M. Ensaf and M. Barati, "Harnessing kernel regression for stochastic state estimation in solar-integrated power grids," arXiv preprint arXiv:2309.10200, 2023, version 1. [Online]. Available: https://arxiv.org/abs/2309.10200

[26] A. M. Tartakovsky, T. Ma, D. A. Barajas-Solano, and R. Tipireddy, "Physics-informed gaussian process regression for states estimation and forecasting in power grids," *International Journal of Forecasting*, vol. 39, no. 2, pp. 967–980, 2023.

[27] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," arXiv preprint arXiv:2604.15686, 2026. [Online]. Available: https://arxiv.org/abs/2604.15686

[28] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.

JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.



MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.



PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.



DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.

Digital Object Identifier

Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]
[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

ABSTRACT This supplement documents the implemented recovery-aware history bank: its local filtering calculations, measurement map, publication rules and simulation contracts. It records the complete confirmatory outcomes, controlled severity-score/state-carry ablation and continuous single-state control. The operating-point signature and recovery-geometry studies keep their separate, known-onset contracts and denominators. All empirical results reuse the frozen evidence; no new campaign or threshold selection is introduced here. Earlier voltage-reconstruction, simultaneous-source and general diagnosability studies are preserved in the separately compiled Historical Technical Notes.

INDEX TERMS phasor measurement units, sequential diagnosis, physical state continuity, conditional filtering, reproducibility.

APPENDIX A IMPLEMENTED PREDICTOR AND CONDITIONAL UNCERTAINTY

This section specifies the local operators and conditional distributions used by the frozen continuous estimator. It distinguishes identities of that local model from approximations introduced by its finite history bank. The numerical checks below exercise these identities on controlled inputs; they do not recalibrate or rerun the reported evaluation.

A. AFFINE OPERATORS AT A CONSISTENT TOPOLOGICAL ANCHOR

For network hypothesis ℓ, let $(\bar{x}_\ell, \bar{u}_\ell)$ be the operating point determined by the prefix-fitted load and dispatch factors. Fix its differential coordinates while solving the algebraic constraints separately for each topology q:

$$g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0. \tag{1}$$

The resulting point need not be a differential equilibrium after a line opening. Assume a nonsingular algebraic Jacobian and a fixed angle gauge. In the expressions below, f is the differential vector field after division by any nonunit differential mass entries. All derivatives are evaluated at the topology-specific point in (1); the network index is suppressed for readability. Define

$$\begin{aligned} c_q &= f_q(\bar{x}, \bar{z}_q, \bar{u}), \\ A_q^c &= f_x - f_z g_z^{-1} g_x, & B_q^c &= f_u - f_z g_z^{-1} g_u, \\ C_q^c &= H_x - H_z g_z^{-1} g_x, & D_q^c &= H_u - H_z g_z^{-1} g_u, \\ d_q^{el} &= H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u}). \end{aligned} \tag{2}$$

For the admitted scalar load direction $\delta u = La$, the continuous approximation is $\dot{\delta x} = A_q^c \delta x + c_q + B_q^c La$ and $\delta y = C_q^c \delta x + d_q^{el} + D_q^c La$. For a line, the topology selects the operator and the amplitude direction is zero. A nonzero pre-event state contributes through δx; d_q^{el} supplies the zero-deviation offset at the fitted anchor. This is a local expansion, so large excursions can still cause prediction error even when the algebraic solve and derivatives are accurate.

For a mode lasting h, compute the affine flow without inverting A_q^c:

$$\mathcal{A}_q = \begin{pmatrix} A_q^c & c_q & B_q^c L \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\exp(h\mathcal{A}_q) = \begin{pmatrix} F_q(h) & \beta_q(h) & U_q(h) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Thus $\beta_q(h) = \int_0^h \exp(A_q^c s) c_q \, ds$. If an admitted switch divides one report into an old-mode interval h_1 and a new-mode interval h_2, then

$$F = F_2 F_1, \qquad \beta = F_2 \beta_1 + \beta_2, \qquad U = F_2 U_1 + U_2. \quad (4)$$

The physical state is continuous across this composition. The final-mode measurement map uses the consistent algebraic right limit. The predictor then appends phase history and the deterministic memory of the instrumental transformation. Scaling these coordinates and applying the same affine augmentation to (4) produces the archived $T_o, J_o, g_o, u_o, j_o, v_o$ operators. In their report ordering,

$$\begin{aligned} r_k &= J_o \xi_k + j_o + v_o a + \epsilon_k, \\ \xi_{k+1} &= T_o \xi_k + g_o + u_o a + \eta_k. \end{aligned} \quad (5)$$

The measurement is scored before correcting the state for the next report. The constant startup discrepancy is applied to the observations; it does not replace the topology-dependent affine terms.

The static builder build_static_rx.jl solves (1) and exports `force`, `pmu`, and their reduced Jacobians; native_conditional_builder.py constructs `offset`, `flow`, and `operator`; `static_bank.py` verifies that the constructed reference matches the prefix-fitted operating point. The reproduction manifest preserves these files and their hashes. No event truth is used to select the startup operating point.

B. FOUR AMPLITUDE INTERVALS REPRESENT ONE PRIOR

Let $I = [-0.2, 0.2]$ be partitioned into four disjoint intervals I_j up to endpoints of zero probability. Write $Z_J(\mu, v) = \int_J \mathcal{N}(a; \mu, v)\, da$ for any interval J. The initial prior has the exact decomposition

$$\begin{aligned} p_0(a) &= \sum_{j=1}^{4} \alpha_{j,0} \, \mathrm{TN}_{I_j}(a; \mu_0, v_0), \\ \alpha_{j,0} &= Z_{I_j}(\mu_0, v_0) / Z_I(\mu_0, v_0). \end{aligned} \quad (6)$$

The implementation assigns one history per source, proposed onset, restoration path, and interval. Each retains its own $(m_{h,j}, d_{h,j}, \mu_{h,j}, v_{h,j}, w_{h,j})$; there is no per-report collapse to a single amplitude distribution. The four interval histories use the same physical operators. They are neither quadrature nodes nor four separately linearized severity models.

For a fixed physical history with observations Y, let $Z_j(Y) = \int_{I_j} p(Y \mid a, h)\, \mathrm{TN}_{I_j}(a; \mu_0, v_0)\, da$. Then

$$p(Y \mid h) = \sum_j \alpha_{j,0} Z_j(Y), \qquad \alpha_j(Y) = \frac{\alpha_{j,0} Z_j(Y)}{p(Y \mid h)}. \quad (7)$$

Applying the local predictive integral sequentially gives the same identity. Identical linear operators and gains preserve the common untruncated Gaussian parameters for interval copies of the same path; their truncated moments and normalized weights differ. Subsequent beam pruning can remove interval histories, so equivalence with an unpartitioned bank is not guaranteed after pruning. The recorded discarded mass describes that approximation and is not a bound on all future decisions.

C. AMPLITUDE DEPENDENCE AND MIXTURE COMPRESSION

Condition on a retained history and the reports used so far. For $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, set $\sigma = \sqrt{v}$, $\alpha = (l-\mu)/\sigma$, $\beta = (u-\mu)/\sigma$, $Z = \Phi(\beta) - \Phi(\alpha)$, and $\lambda = [\phi(\alpha) - \phi(\beta)]/Z$. Its moments are

$$\begin{aligned} \bar{a} &= \mu + \sigma\lambda, \\ V_a &= v \left[1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z} - \lambda^2 \right]. \end{aligned} \quad (8)$$

The frozen `trunc` kernel evaluates the normalizer in the log domain and bounds small numerical excursions of the moments. The Gaussian parameters μ, v are therefore distinct from the conditional amplitude mean and variance.

For the affine conditional state model $\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_h)$, total expectation and covariance give

$$\begin{aligned} \bar{\xi}_h &= m_h + d_h \bar{a}_h, \\ \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ V_h &= P_h + d_h V_{a,h} d_h^\top. \end{aligned} \quad (9)$$

With normalized history weights ω_h, including the no-new-event history, the complete mixture moments would be

$$\begin{aligned} \bar{\xi} &= \sum_h \omega_h \bar{\xi}_h, \\ P_{\mathrm{mix}} &= \sum_h \omega_h \left[V_h + (\bar{\xi}_h - \bar{\xi})(\bar{\xi}_h - \bar{\xi})^\top \right]. \end{aligned} \quad (10)$$

This identity applies within a network hypothesis and its common state coordinates. It does not justify averaging absolute rotor states between different network equilibria.

The continuous estimator uses the shared regime covariance P_ρ in place of P_h. It updates the affine means and amplitude distributions, and computes marginal diagonal state variances for diagnostics. At closure it carries $\bar{\xi}$, clears explicit histories, and records the trace of amplitude/mean spread omitted from the new single-history representation. It does not carry the complete covariance in (10), nor the uncertainty of the fitted startup context. Instrumental and discrepancy states remain in the carried augmented mean.

For comparison, an exact conditional covariance recursion for the local linear model (5), with independent noises of covariances R, Q, is

$$\begin{aligned} S_h &= J_o P_h J_o^\top + R, \\ K_h &= T_o P_h J_o^\top S_h^{-1}, \\ P_h^+ &= T_o P_h T_o^\top + Q - K_h S_h K_h^\top. \end{aligned} \quad (11)$$

This is a diagnostic reference, not a description of a new executed continuous estimator. The frozen implementation solves the Riccati equation for the nominal operator in each

TABLE 1. Deterministic local-model checks. Maximum absolute errors over 72 scalar updates, five switch positions, and 12 joint Gaussian updates; each row uses its stated reference.

Check	Maximum error
Log evidence: partition	1.78×10^{-15}
Amplitude mean/variance	4.16×10^{-17}
Log evidence: quadrature	3.11×10^{-15}
Affine flow: ODE reference	1.11×10^{-16}
Joint mean/covariance	8.33×10^{-17}

discrepancy regime and uses that shared covariance for the admitted switched operators.

Sequential likelihood factorization conditions on past reports; it does not require marginal independence of the observations. The instrumental VAR/AR filter and the latent discrepancy AR state already model temporal dependence. Remaining covariance misspecification, linearization error, adaptive proposals, and compression can nevertheless affect the resulting weights. Their empirical calibration requires pre-update residual and coverage diagnostics, separately from the algebraic checks below.

D. DETERMINISTIC VERIFICATION AND ITS BOUNDARY

`scripts/verify_history_partition.py` loads the frozen amplitude kernel without running a campaign. It compares sequential one-interval and four-interval evidence and moments with direct scalar quadrature, including zero information and amplitudes on interval boundaries. An independent small switched linear system checks affine-flow composition against numerical integration. A linear Gaussian reference also compares the conditional affine-state recursion with a joint state–amplitude Kalman recursion for an untruncated Gaussian amplitude; the truncated case is checked separately by quadrature. All histories are retained in these checks. Source hashes, tolerances, and complete results accompany the generated table. These checks establish local-model identities on the stated inputs. They establish neither calibration of the pruned network bank nor robustness of its nonlinear physical approximation, and add no event outcomes to the evaluation cohorts.

APPENDIX B HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections B–D give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the frozen method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the frozen operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The frozen screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode

closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.

7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the frozen detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{12}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FROZEN NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their frozen publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j / \max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{13}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a frozen proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX C LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \qquad (14)\end{aligned}$$

$$v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \qquad (15)$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \qquad (16)\end{aligned}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \qquad (17)$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (16) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. The script scripts/verify_supplement_examples.py checks the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a frozen experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned}S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \qquad (18)\end{aligned}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa}) e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \qquad (19)$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \qquad (20)\end{aligned}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX D SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The active simulator is the frozen research candidate `SIM0001_JuliaPD_F01`, not an officially validated equivalent of RAW0001. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the frozen impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a

nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. RAW0001 provided retrospective calibration context. RAW0002 remains outside these studies.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

H03 and H04 each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. Chen2016 is eligible on that lane; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The E_i/X02 signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The X04 recovery geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's frozen plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the F01 state, parameter and measurement files from its snapshot. For any new simulation create a new run directory; a new hypothesis requires a successor candidate. Do not run historical preparation scripts inside a frozen baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures with scripts/build_manuscript_bundle.ps1. Its default compilation does not regenerate scientific artifacts or import new campaign results. An explicit artifact refresh is a separate operation.

The repository keeps the current walkthrough in this appendix, the frozen campaign evidence under `generated/frozen/`, and the earlier mathematical development in the separately compiled Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX E EXECUTED MEASUREMENT MAP

A. F01 MEASUREMENT CONTRACT

The executed channel map is taken from the frozen SIM0001-JuliaPD-F01 research snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral RMS volts using $345\,\text{kV}/\sqrt{3}$; voltage angle uses one common affine phase reference plus the archived relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current

bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the frozen stream. The stochastic model archives a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

APPENDIX F LEARNED COMPARATOR LINEAGE AND DECISION CONTRACT

Both models were fitted on 32 physical parents from the four earlier development campaigns `SIM0014–SIM0017`; validation used 8 parents and 16 observed records from `SIM0018`. These identifiers name historical campaigns that were already development data for this comparator construction. The fitted label set contains normal operation and all 18 catalog sources. The training target is the active source; after restoration, including physical recovery, its label is normal. This is not a source-excluded transfer test.

The confirmatory loader selects the two `M03_R09` artifacts in Table 3, not the original hackathon serializations. Each report selects the largest tree vote, and six consecutive reports of the same label drive the episode interface. The wrapper has no separate probability-trigger threshold to lower. Tree votes are uncalibrated; the 300-report prefix and monitoring start at zero-based report 360 are frozen.

Full model and source hashes, training metadata and the observed class set are recorded in generated/comparator_lineage.json. The reproduction snapshot is generated/frozen/comparator_lineage_20261103/. Running scripts/build_comparator_lineage.py regenerates this table from that snapshot without the research checkout, model fitting, predictions or campaign outputs.

TABLE 2. Eight PMU rows in the frozen F01 measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	F01 archived terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	F01 archived terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	F01 archived terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	F01 archived terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	F01 archived terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	F01 archived terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

TABLE 3. Identity of the fitted ExtraTrees comparators used in the confirmatory streams. Hashes show the first 12 hexadecimal characters; the frozen provenance JSON retains all 64. These are development-trained artifacts, distinct from the original submitted hackathon models.

Comparator	Artifact variant	Features	Trees	Model SHA-256 prefix
ExtraTrees	BASELINE	846	400	`8be07d191ec9`
ExtraTrees short	SHORT	1457	400	`e4afa2907d5f`

APPENDIX G COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The frozen test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the sealed summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 1 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the frozen first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective RAW0001 replay under its startup noise-scale contract.

TABLE 4. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 5. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the frozen evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen2016
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 8. Persistent-line coverage and stopping outcomes across six repeated strata, including both frozen ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen2016 normal exposure ends at its first stop; the other methods monitor the full available record.

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen2016	0	74/162	66	0	72/72	0.701

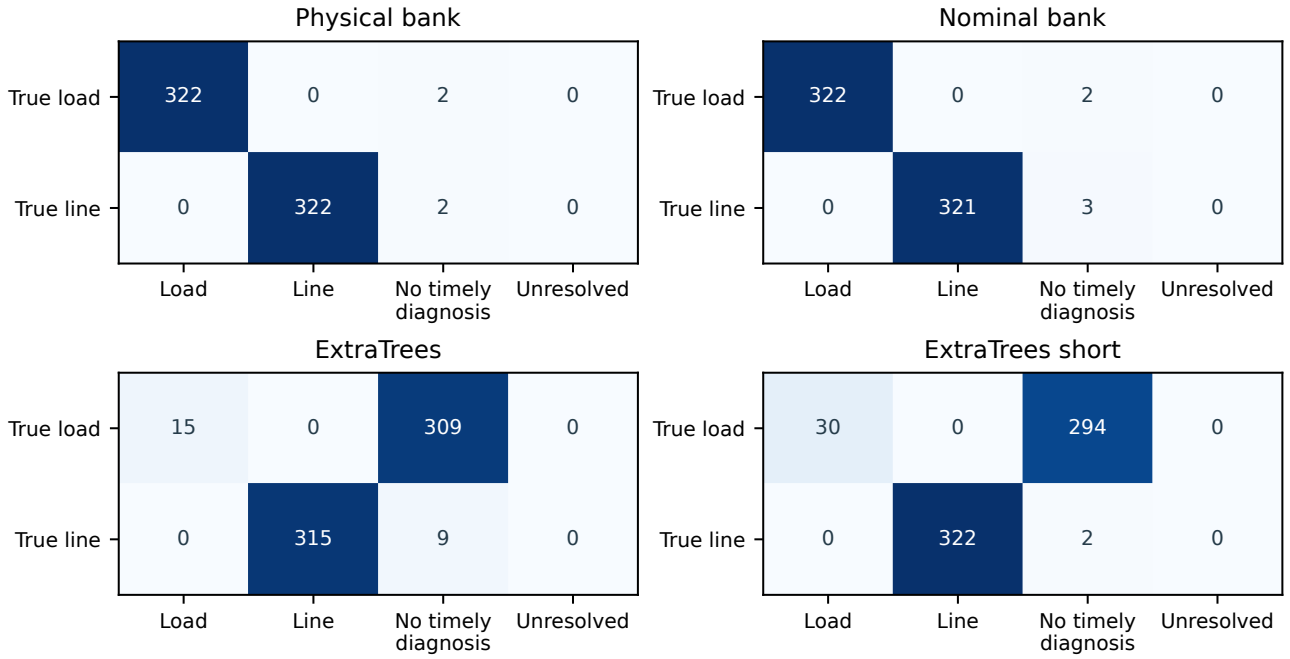


FIGURE 1. **Posthoc confusion audit of the frozen continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the prediction seal. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 9. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 10. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain archived.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-7.41 [-16.67, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-7.41 [-14.81, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-7.41 [-14.81, -1.85]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-7.41 [-14.81, -1.85]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.56 [-14.81, 5.56]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.56 [-14.81, 5.56]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.70 [-12.96, 7.41]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.70 [-12.96, 5.56]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.85 [0.00, 5.56]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.56 [-11.11, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.70 [0.00, 9.26]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.70 [-11.11, 3.70]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	59.26 [40.74, 79.63]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	61.11 [40.74, 81.48]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	75.93 [51.85, 100.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	77.78 [50.00, 105.56]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	55.56 [38.89, 72.22]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	46.30 [27.78, 66.67]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	72.22 [46.30, 98.15]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	62.96 [35.19, 88.89]

TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX H CONTROLLED COMPONENT ABLATION

The frozen protocol and all output tables are supplied in the reproduction archive under generated/frozen/ablation_a01_20261102. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

a: Exposure convention for the archived ablation rate. The ablation table retains the frozen metric extras_per_event_hour, whose denominator is the complete 120-s event record, including startup. Its continuous exposure is $18 \times 120 = 2160$ s. The main article's confirmatory alarm-rate figure instead uses monitored exposure after the 12-s startup. Under that convention the ablation exposure would be $18 \times 108 = 1944$ s and each reported rate difference and interval bound would be multiplied by $120/108$. Counts, event matches and the archived table are unchanged; these two rate conventions must not be compared without this conversion.

APPENDIX I EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - a s_k$. The affine conditional state is $m_k + a d_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{21}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{22}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{23}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a} U - \tfrac{1}{2} \widehat{a}^2 G - C_{ref}. \tag{24}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c) P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying

TABLE 13. All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.

method	lane	available	events	timely	extras	false restorations	open eof
Marginal + carry	continuous	18	108	108	1	1	0
Marginal + carry	normal	3	0	0	0	0	0
Marginal + carry	persistent_line	3	3	3	0	0	3
Marginal + reset	continuous	18	108	75	166	130	11
Marginal + reset	normal	3	0	0	0	0	0
Marginal + reset	persistent_line	3	3	3	0	0	3
Profile + carry	continuous	18	108	108	1	1	0
Profile + carry	normal	3	0	0	0	0	0
Profile + carry	persistent_line	3	3	3	0	0	3
Profile + reset	continuous	18	108	75	166	130	11
Profile + reset	normal	3	0	0	0	0	0
Profile + reset	persistent_line	3	3	3	0	0	3

TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.

method	reference	metric	difference	lower95	upper95
Marginal + carry	Profile + carry	timely_fraction	0.000	0.000	0.000
Marginal + carry	Profile + carry	extras_per_event_hour	0.000	0.000	0.000
Marginal + carry	Marginal + reset	timely_fraction	0.306	0.241	0.380
Marginal + carry	Marginal + reset	extras_per_event_hour	-275.000	-325.000	-223.333
Marginal + reset	Profile + reset	timely_fraction	0.000	0.000	0.000
Marginal + reset	Profile + reset	extras_per_event_hour	0.000	0.000	0.000
Profile + carry	Profile + reset	timely_fraction	0.306	0.241	0.380
Profile + carry	Profile + reset	extras_per_event_hour	-275.000	-325.000	-223.333

TABLE 15. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The H04 comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

TABLE 16. What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

APPENDIX J DEVELOPMENT STREAMS AND ADMISSION CONTROL

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 16). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multiclass hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 3 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

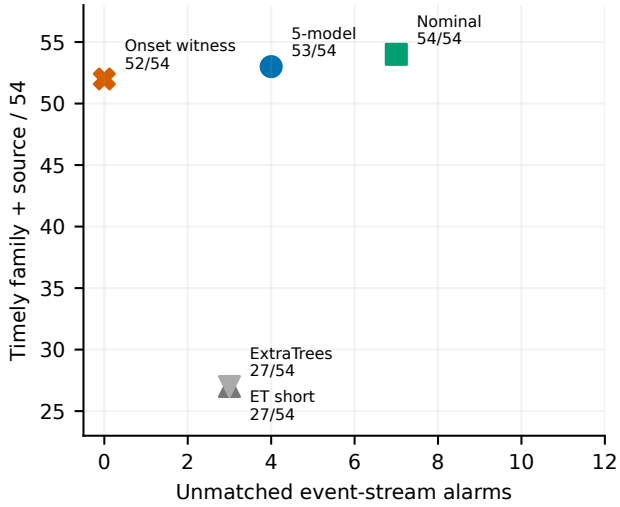


FIGURE 2. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative experiment leaves R11 and nominal frozen for C01.

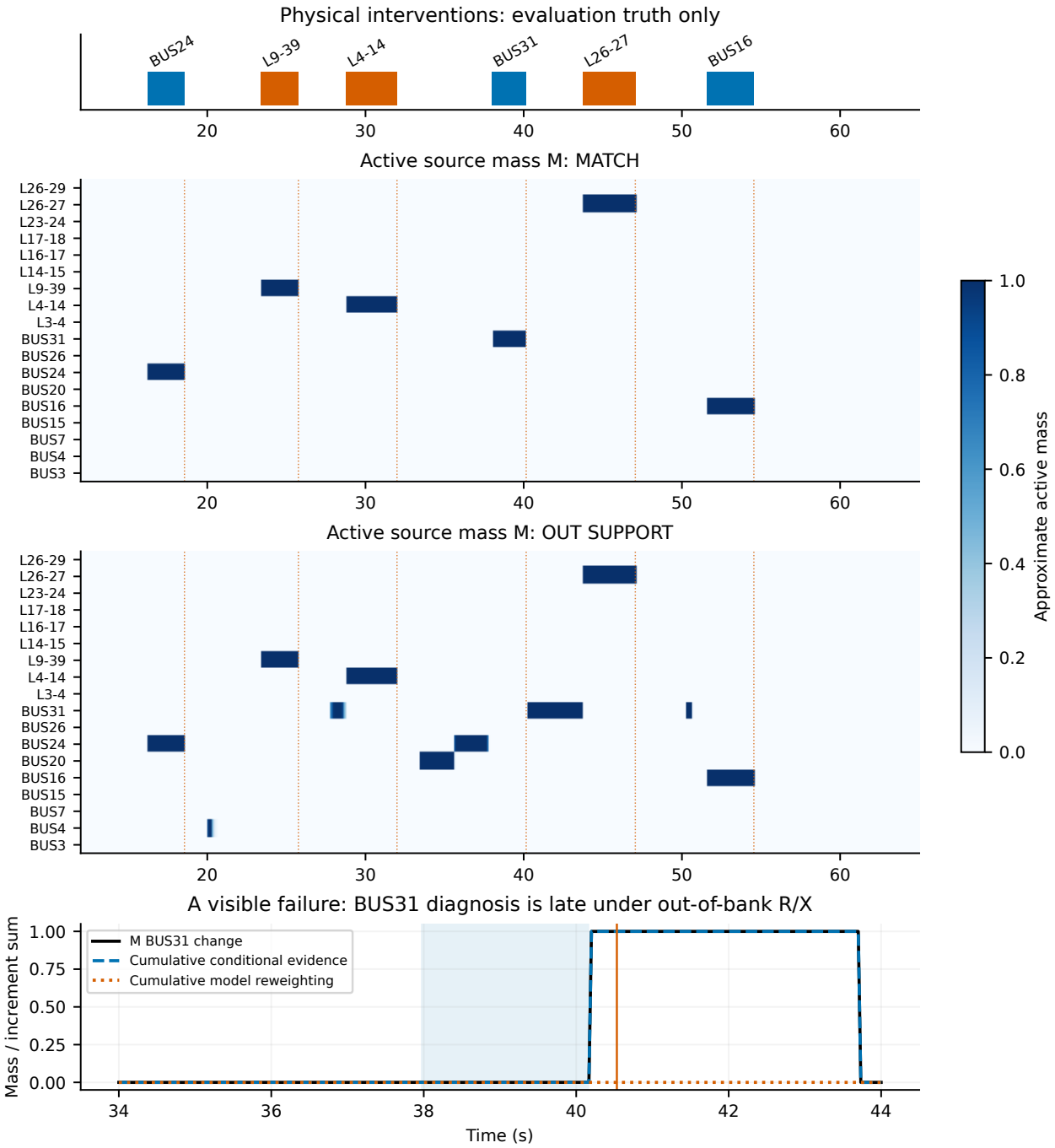


FIGURE 3. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

APPENDIX K OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of R11.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{25}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{26}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 5 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

APPENDIX L RECOVERY GEOMETRY AND SOURCE SEPARATION

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new

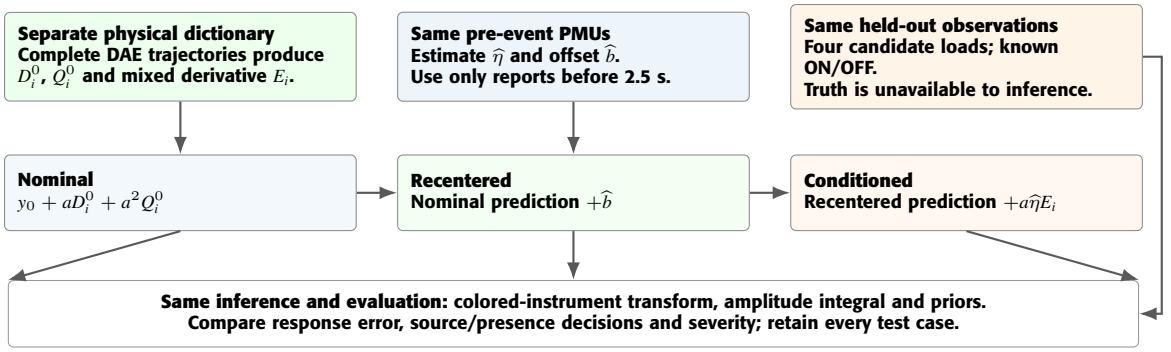


FIGURE 4. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

TABLE 17. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

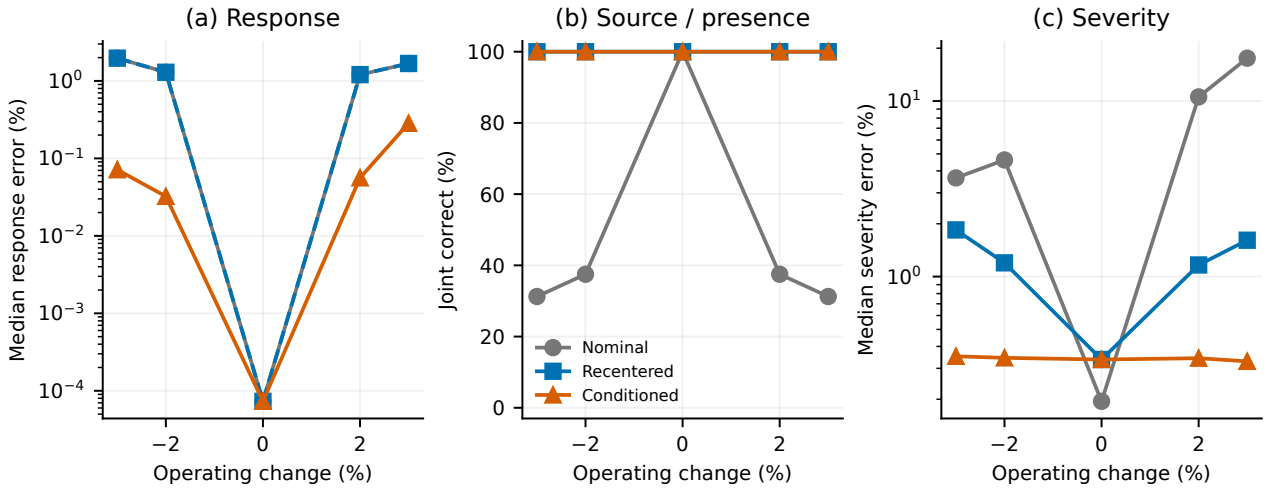


FIGURE 5. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{27}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{28}$$

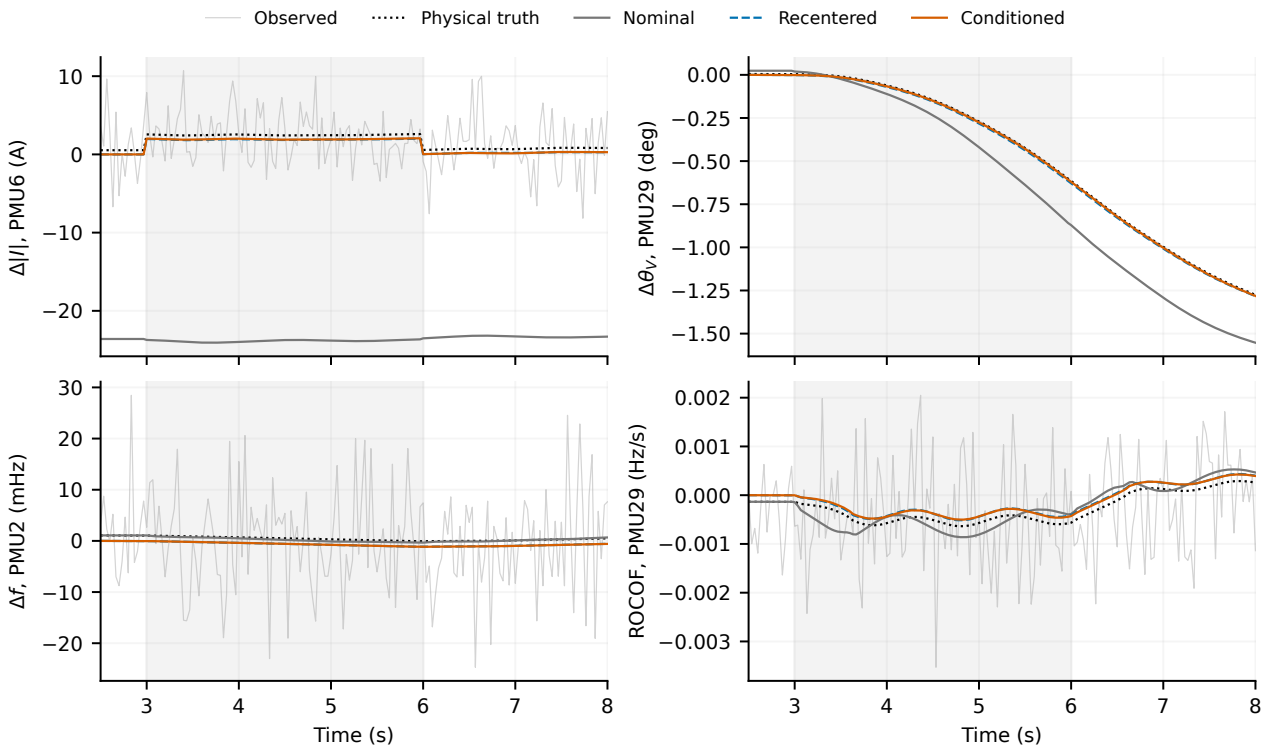


FIGURE 6. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [2], [3].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 7). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation

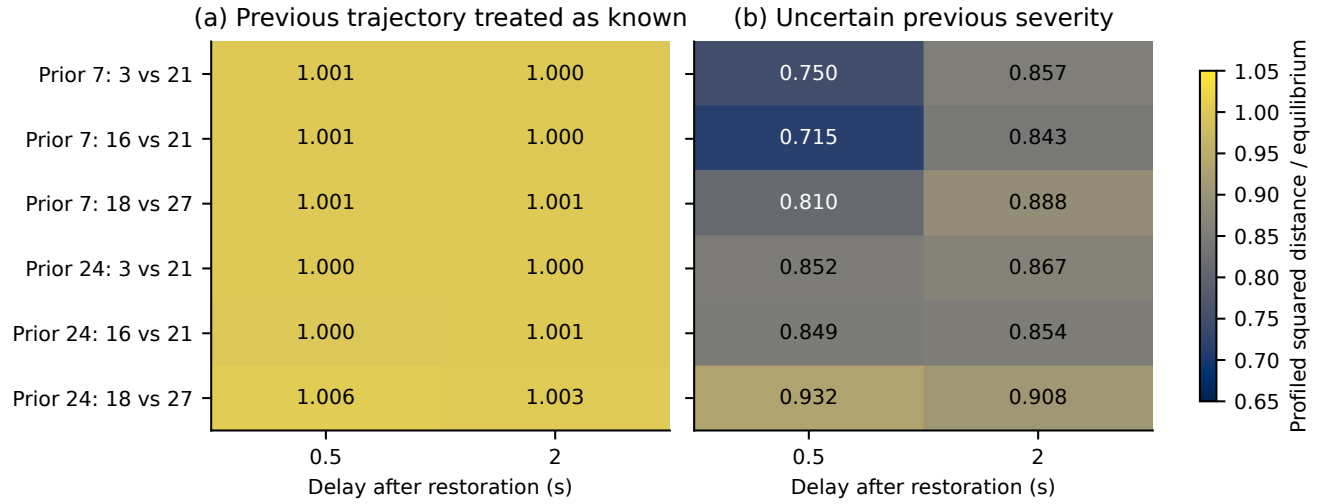


FIGURE 7. **Full-F01 physical control, not an R11 performance comparison. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an R11 improvement nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.

[2] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[3] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.